ANIRUDH SARVE

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate specializing in Applied Machine Learning, Computer Vision, Deep Sensor Fusion, and High-Performance Distributed Systems. Proven track record in formulating physics-constrained Kalman filters, cross-modal remote sensing reconstruction networks, and edge neural runtimes (ONNX, WebAssembly, ARM Cortex-A) delivering mathematically bounded stability and sub-3 ms latency budgets. Lead Architect for the ISRO Smart India Hackathon (SIH 2026), bridging theoretical AI research with production-grade engineering.

EDUCATION

Vidyalankar Institute of Technology (VIT), Mumbai — **B.E. in Computer Engineering** **Expected May 2028**
Cumulative GPA: 8.90 / 10.0 | Affiliated with University of Mumbai | **Class XII (HSC): 92.60%** (Distinction)
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Vision, Sensor Fusion, Operating Systems, Database Management Systems.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript (Node.js), SQL, Go (Basics), HTML5/CSS3.
AI & Machine Learning: PyTorch, TensorFlow, ONNX, OpenCV, Faster-Whisper, Scikit-learn, 15-State ESKF, Sensor Fusion, ResU-Net, LangGraph.
Web & Systems: Next.js 16, React 19, FastAPI, Express.js, WebAssembly (WASM), WebGL, WebSockets, Docker, Linux, Git.
Databases & Cloud: PostgreSQL, MongoDB, Upstash Redis (Distributed Locks), Qdrant Vector DB, SQLite (FTS5 BM25), AWS, GCP.

PROFESSIONAL EXPERIENCE

NxtChaptr — **Software Developer (Edge AI & Full-Stack Infrastructure)** **Nov 2024 – Aug 2025**

- Architected client-side edge computer vision using U²-Net via **WebAssembly/WebGL**, processing image photo-matting in **<3.5s** and eliminating recurring server GPU costs.
- Engineered automated catalog metadata extraction microservices using **Google Gemini Vision**, reducing merchant listing creation latency from 8 minutes to **<30 seconds**.
- Integrated Upstash Redis atomic distributed locks (SET NX EX) and Ably WebSockets to eliminate race-condition overselling across concurrent 1-of-1 product drops.

Wroomie — **Software Developer (Backend & Applied ML)** **Nov 2023 – Jan 2024**

- Developed multi-attribute compatibility scoring algorithms clustering user lifestyle preferences across 4 behavioral vectors, achieving **94% match satisfaction**.
- Optimized MongoDB geospatial room discovery queries with 2dsphere indexing, decreasing query latency by **35%** across high-density university zones.

FEATURED TECHNICAL PROJECTS

Neural-Inertial Dead-Reckoning Engine with 15-State Error-State Kalman Filter (ISRO SIH PS 26168)
Technologies: PyTorch, 15-State ESKF, 5-Head Temporal Attention GRU, ONNX, C++, ARM Cortex-A, ROS2

- Engineered hybrid dead-reckoning engine holding vehicle position drift to **≤ 1.85 m** across 120s GPS outages (a **98.4% drift reduction** versus standard EKF baselines).
- Fused 10 Hz raw IMU with continuous-time 15-state ESKF, enforcing physical Non-Holonomic Constraints (NHC) and ZUPT updates at sub-3 ms CPU latency on ARM Cortex-A ECUs.

Multimodal Optical-SAR Fusion ResU-Net with Cross-Modal Attention (ISRO LISS-IV)
Technologies: PyTorch, ResU-Net, Cross-Modal Attention, Sentinel-1 SAR (C-Band), GeoTIFF, GDAL

- Constructed generative AI framework for cloud removal in ISRO Resourcesat-2/2A LISS-IV satellite imagery (5.0 m resolution), fusing optical and Sentinel-1 SAR radar backscatter.
- Formulated differentiable Spectral Angle Mapper (SAM) loss, achieving **32.09 dB PSNR**, **0.7167 SSIM**, and tight **6.17° SAM angle** on real validation scenes.

Neuro-Symbolic Hybrid Retrieval Engine & Automated Standards Compliance (SIH PS 26108)
Technologies: React 18, TypeScript, FastAPI, SQLite FTS5 BM25, ONNX Dense Embeddings, PaddleOCR v2.9+

- Built 3-way hybrid retrieval engine (SQLite FTS5 BM25 + regex + dense embeddings) via Reciprocal Rank Fusion across 19,423 BIS engineering standards.
- Integrated verification kernel with PaddleOCR pipeline achieving **99.28% OCR confidence** on complex multi-column tender BoQ tables.

Sub-150ms Vector Graphics Stream PDF Compiler & Model Context Protocol (MCP) Server
Technologies: TypeScript, Turborepo, Rust Canvas Bindings, Zod, MCP Protocol, Node.js

- Engineered vector PDF streaming engine compiling live schemas in **<150 ms (<40 MB RAM)**; built native Model Context Protocol (MCP) server for autonomous AI agent tool execution.

KEY ACHIEVEMENTS & AWARDS

- **Lead Architect, ISRO Smart India Hackathon (SIH 2026):** Authored AI-ESKF sensor-fusion architecture for Problem Statement 26168.
- **Autonomous Navigation Team, Indian Rover Challenge (IRC):** Designing kinematic guidance telemetry and sensor-fusion simulation pipelines.
- **Scholastic Honors:** 8.90 / 10.0 CGPA at VIT Mumbai; 92.60% (Distinction) in Maharashtra State Board HSC Examination.
- **Open Source Contributor & Architect:** Published vector PDF streaming compilers, MCP toolkits, and computer vision utilities on GitHub.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Autonomous Systems Subsystem Lead, VIT Robotics Club:** Led telemetry and ROS2 navigation stack development for collegiate robotics competitions.
- **Technical Workshop Speaker:** Conducted hands-on technical sessions on Edge AI optimization and WebAssembly for 200+ engineering students.

HOBBIES & PERSONAL INTERESTS

Technical & Creative Pursuits: Embedded robotics tinkering (ROS2 / micro-ROS), Autonomous drone navigation, Competitive chess (Tactical positional play), Amateur astronomy & satellite orbit tracking, Technical writing & systems architecture deep-dives.